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Dynamic privacy budget allocation for enhanced differential privacy in federated learning

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Abstract

Traditional applications of differential privacy (DP) in federated learning (FL) usually use a fixed privacy budget. However, this approach has significant limitations when handling data with varying sensitivity between different nodes. This article introduces a novel dynamic privacy budget allocation (DPBA) algorithm to address these shortcomings and improve the effectiveness of DP in FL. The DPBA algorithm considers factors such as node data sensitivity, participation in budget allocation, participation frequency, and the global model convergence state. By dynamically adjusting the privacy budget, DPBA not only increases the flexibility of privacy protection, but also enhances efficiency within the FL framework. This is achieved by allocating the privacy budget based on the data characteristics of each node and the training progress of the model, thus providing stronger protection for sensitive data while minimizing privacy leakage and performance degradation. This paper details the design principles and implementation steps of the DPBA algorithm and validates its applicability and effectiveness through extensive experiments on various datasets. The results show that DPBA not only provides robust privacy guarantees for individual nodes but also substantially improves classification accuracy compared to a fixed-budget approach. These findings underscore the promise of DPBA for federated learning and offer new insights into applying differential privacy in increasingly complex data settings.

Keywords Federated learning · Differential privacy · Dynamic privacy budget · Optimization strategies

1 Introduction

With the rapid advancement of big data and machine learning technologies, data privacy and security have emerged as critical concerns [1, 2]. In this increasingly data-driven era, the risk of personal information leakage and misuse is growing. Ensuring effective data utilization while protecting user privacy has become a major challenge for academia and industry. FL, a distributed machine learning framework, provides some privacy protection by enabling collaborative model training across multiple nodes without centralizing data [3]. However, privacy risks still exist in FL, particularly in complex environments that involve numerous nodes. The exchange of data and model updates between these nodes can potentially expose sensitive information [4, 5].

To address privacy concerns, DP has become a widely adopted technique for protecting data privacy [6, 7]. DP mitigates the risk of identifying individual data records by introducing random noise into data processing, thus providing a robust safeguard for user privacy [8, 9]. However, in the context of FL, the traditional method of using a fixed privacy budget lacks efficiency and flexibility due to significant variations in factors such as data sensitivity, participation frequency, and the state of convergence of the global model between different nodes [10, 11]. This static allocation method often fails to meet the privacy needs of individual nodes, resulting in excessive or insufficient privacy budgets, which can negatively impact both overall model performance and privacy protection [12]. As a result, designing a mechanism that dynamically adjusts the privacy budget has emerged as a critical area of research in FL.

Although DP is widely recognized for its effectiveness in preventing personal data leakage by adding noise to statistical output [13], its application in FL presents several challenges. First, traditional DP methods typically use a fixed privacy budget, which is ill-suited to handle the varying sensitivities of data between different nodes, leading to ineffective privacy protection. Second, a fixed budget can result in uneven privacy protection during the FL process, affecting both model performance and training efficiency. In addition, a central challenge in DP is

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finding the optimal balance between privacy protection and data utility. Managing and allocating the privacy budget effectively in multiple queries is the key to maintaining this balance [14].

The DPBA algorithm flexibly tailors privacy budgets to the needs of each node while simultaneously boosting the learning system's overall efficiency and effectiveness. To address existing challenges, this article introduces DPBA, a novel algorithm designed to optimize the application of DP in FL environments. DPBA treats privacy budget allocation as a dynamic process that intelligently adapts to each node's unique requirements. This process includes an initial setup, an evaluation of data sensitivity for each node, and the calculation and adjustment of privacy budgets based on node participation frequency and the model's convergence state. These dynamically adjusted privacy budgets are applied during model training and update. The strength of the DPBA algorithm lies in its flexibility, which allows it to effectively accommodate varying privacy needs across nodes while significantly improving system-wide learning efficiency and performance. By enabling adaptive privacy budget allocation, DPBA protects sensitive data without compromising model accuracy or convergence speed. This demonstrates its substantial potential for practical applications.

The main contributions of this study are as follows.

(1) This paper introduces a novel DPBA, which enhances traditional privacy budget control mechanisms by incorporating key factors such as data sensitivity, node participation frequency, and model convergence state. This innovation enables personalized privacy budget adjustments based on the specific characteristics of each participating node, providing stronger protection for highly sensitive data while avoiding the overprotection of low-sensitivity data. This approach improves the overall efficiency of privacy protection in FL.
(2) The effectiveness of the DPBA algorithm is validated across different data types and FL environments through experiments on two widely used benchmark datasets, MNIST and CIFAR-10. The experimental results demonstrate that the risk of privacy leakage is effectively reduced by DPBA while model performance is maintained, and that it outperforms traditional methods in various scenarios.
(3) By introducing and validating the DPBA algorithm, this paper extends the privacy protection methodologies in FL and offers a practical solution to allocate the privacy budget in real-world applications. In particular, the study evaluates the adaptability of the algorithm in the cases of uneven data distribution and finds that DPBA significantly improves the model convergence speed, achieving a balance between privacy protection and model accuracy.

The paper is organized as follows: Section 2 introduces the existing studies of FL and DP; Section 3 provides a detailed explanation of the DPBA algorithm; Section 4 presents experimental comparisons using the MNIST and CIFAR-10 datasets, along with parameter analysis; finally, Section 5 summarizes the study and outlines future research directions.

2 Related work

2.1 Federated learning and differential privacy preservation

The evolution of FL and the application of DP are critical topics in contemporary data science and privacy protection. FL enables multiple nodes (for example, smartphones or other devices) to collaborate in training a shared machine learning model while keeping their data local, avoiding the need to upload data to a centralized server [15, 16]. With growing concerns about data privacy and the enforcement of data protection regulations, FL is emerging as a key tool to protect privacy [17]. DP, which provides strong privacy guarantees by adding random noise to the data analysis, ensures that the inclusion or exclusion of any individual in the dataset does not significantly affect the results of the query, thus protecting individual privacy [18, 19]. In FL, DP is used primarily to secure model updates. After local training at each node, noise is added to the updates before they are shared. However, a major challenge is to balance model performance with privacy, as excessive noise can degrade accuracy. Yang et al. [20] introduced FedIPR, a novel ownership verification scheme that embeds watermarks into federated models to protect intellectual property. Khan et al. [21] explored the various domains of FL, highlighting its potential applications in improving the efficiency and flexibility of industrial processes. Their work also discusses the impact of FL on the revolutionization of model training trends in terms of data privacy, decentralization, security, and resource management. Li et al. [22] proposed federated deep reinforcement learning (FedDRL), a scheme that combines FL with deep reinforcement learning for ultra-short-term wind power forecasting. Wang et al. [23] presented a secure data circulation and sharing scheme for industrial IoT (I-IoT) devices, using FL on a trusted implementation platform.

2.2 Privacy budget management

The rational allocation and management of privacy budgets throughout the FL process is crucial to maintaining both learning efficiency and privacy protection. In DP, the privacy budget quantifies the risk of revealing personal information during data analysis [24]. It plays a central role in the balance between data utility and privacy protection, serving as a key tool for designing and evaluating DP mechanisms. A lower privacy budget indicates stronger privacy protection, as it involves adding more

noise to the data. Traditionally, static privacy budget allocation methods assign a fixed value before data analysis begins. Although this approach is simple and easy to implement, it may not always be optimal. Friha et al. [25] proposed a secure and decentralized approach that incorporates a differentially private gradient exchange scheme to secure smart industrial facilities. Shi et al. [26] introduced DP-FedSAM, a novel differentially private algorithm that utilizes gradient perturbations to mitigate the adverse effects of DP. DP-FedSAM integrates a sharpness-aware minimization optimizer, producing locally flat models that are more stable and robust against weight perturbations, thereby achieving smaller norms for local updates and greater resistance to noise. So et al. [27] developed Multi-RoundSecAgg, a secure aggregation framework with multiple rounds of privacy guarantees. They introduced a new metric to quantify privacy in multi-round training and devised a structured user selection strategy to ensure long-term privacy for each user. Wang et al. [28] proposed a mechanism that combines DP with reinforcement learning, where the randomization property of DP mitigates the impact of malicious agents, and reinforcement learning allows agents to make decisions in social networks. This mechanism promotes cooperation in adversarial environments. Zhao et al. [29] addressed the multivariate prediction problem of human mobility using a differentially private deep learning model that uses gradient perturbations using a differentially private stochastic gradient descent algorithm to protect each time series sample. Input perturbations are also applied, ensuring privacy at the sample level before learning begins.

Static approaches fail to account for the unique characteristics of datasets and the dynamic nature of the analysis tasks. In practice, privacy requirements can vary significantly across datasets and over time. Fixed privacy budgets lack the flexibility to adapt to these changes, often leading to excessive or inadequate privacy protection. In FL, participating nodes can handle datasets with varying sensitivities and privacy needs. Using dynamic privacy budgets, privacy protection can be tailored to the unique characteristics of each node, improving the overall effectiveness of privacy safeguards. In this paper, we propose the use of dynamically adjusted privacy budgets to better manage privacy risks throughout different stages of the learning process, ensuring continuous and consistent privacy protection.

3 Methodology

3.1 Symbols, assumptions, and task definitions

In the DPBA method, N represents the total number of nodes in the FL system, where i is the index of a node ($i \in \{1, 2, \ldots, N\}$), and D_i is the dataset owned by node i. The total privacy budget of the system is indicated by ϵ_{total}, while ϵ_i is the privacy budget allocated to node i, and t is the time step or iteration number. Sens_i refers to the sensitivity of the data of node i, $\text{Freq}_i(t)$ is the frequency of participation of node i at time t, and $\text{Conv}_i(t)$ is the contribution of node i to the convergence of the model at time t. Parameters α and β are tuning factors that balance the influence of different factors on the allocation of privacy budget.

In this paper, the DPBA algorithm is designed based on the following assumptions: the distribution of each node's dataset D_i is relatively stable during the iteration process, which can be accurately quantified by the sensitivity scoring function Sens_i; the node's frequency of participation at each iteration t $\text{Freq}_i(t)$ is highly correlated with the actual contribution and can be accessed in real time; the node's influence on the global model convergence can be approximated by the metric function $\text{Conv}_i(t)$; the total privacy budget of the system ϵ_{total} is kept fixed throughout the federated learning process, and the privacy budget $\epsilon_i(t)$ allocated to each node needs to satisfy the requirements of $\sum_{i=1}^{N} \epsilon_i(t) \leq \epsilon_{\text{total}}$; the weighting parameters α, β satisfy $\alpha + \beta = 1$,which are used to balance the data sensitivity with the contribution to convergence; when $\sum_i \epsilon_i(t) > epsilon_{\text{total}}$, proportional scaling is used to ensure the total budget constraint.

In an FL environment, the objective of the DPBA algorithm is to dynamically adjust the privacy budget ϵ_i allocated to each node based on its specific characteristics and requirements, while ensuring that the system meets general privacy protection standards and maintains learning efficiency. This approach addresses the limitations of traditional fixed privacy budget methods in handling data diversity across nodes and enhances both the learning efficiency and privacy protection of the entire system.

3.2 DPBA framework

The DPBA algorithm is specifically designed to adapt to the diversity and dynamism of FL environments. It achieves more efficient privacy protection by flexibly adjusting the allocation of the privacy budget while ensuring that the performance of the FL model remains uncompromised. For each node i in the FL system, the privacy budget $\epsilon_i(t)$ is dynamically calculated and allocated based on the sensitivity Sens_i of its dataset D_i, its participation frequency $\text{Freq}_i(t)$ at time t and its contribution to model convergence $\text{Conv}_i(t)$ at time t. Gleichzeitig, it ensures that the total privacy budget across all nodes at any time t does not exceed the predetermined general privacy budget ϵ_{total}.

By balancing $\epsilon_i(t)$ and optimizing the overall learning efficiency, the DPBA algorithm achieves an optimal trade-off between privacy and efficiency through the tuning parameters α and β, where $\alpha + \beta = 1$. The framework of

the DPBA algorithm is illustrated in Fig. 1. As shown in Fig. 1, the algorithm comprises the following five steps:

3.2.1 System model

In the system model, we consider an FL system comprising a trusted central server S and N geographically distributed user nodes $\{U_i\}_{i=1}^{N}$. Each node U_i locally stores a private dataset D_i containing $|D_i|$ samples, and at the beginning of the communication round t has a synchronized copy of the global model parameters $W^{(t)}$. To enforce end-to-end differential privacy, the server defines a global privacy budget ϵ_{total}, which represents the maximum allowable privacy expenditure on all rounds. At the start of each round, the server divides this budget among the nodes, assigning to node U_i a per-round budget $\epsilon_i(t)$, subject to the constraint:

$$\sum_{i=1}^{N} \epsilon_i(t) \;\le\; \epsilon_{\text{total}}. \tag{1}$$

Upon receiving $W^{(t)}$, node U_i computes its empirical risk on its local data,

$$\ell_i\big(W; D_i\big) = \frac{1}{|D_i|} \sum_{(x,y)\in D_i} \mathcal{L}\big(f_W(x),\, y\big), \tag{2}$$

and then performs a single gradient descent step with noise calibrated to $\epsilon_i(t)$. Specifically, it first evaluates the gradient $\nabla \ell_i\big(W^{(t)}; D_i\big)$, whose sensitivity is indicated by Δf, and samples noise from a Laplace distribution of the scale parameter $\Delta f/\epsilon_i(t)$. The private update is thus

$$W_i^{(t+1)} = W^{(t)} \;-\; \eta \left(\nabla \ell_i\big(W^{(t)}; D_i\big) + \text{Laplace}\big(0, \tfrac{\Delta f}{\epsilon_i(t)}\big)\right), \tag{3}$$

where η is the learning rate. This step ensures that the contribution of U_i to the global model respects $\epsilon_i(t)$-differential privacy.

Once noisy updates $\{W_i^{(t+1)}\}_{i=1}^{N}$ are returned, the server aggregates them via a weighted average, with weights proportional to each dataset size:

$$W^{(t+1)} = \sum_{i=1}^{N} \frac{|D_i|}{\sum_{j=1}^{N} |D_j|}\, W_i^{(t+1)}. \tag{4}$$

This aggregation step produces the next global model, balancing contributions from nodes with differing data volumes.

Finally, before starting the round $t+1$, the server recalculates each node's privacy budget $\epsilon_i(t+1)$ by blending three node-specific metrics:

$$\epsilon_i(t+1) = \alpha\, \text{Sens}_i\, \frac{\text{Freq}_i(t)}{\sum_{j=1}^{N} \text{Freq}_j(t)} \;+\; \beta\, \text{Conv}_i(t), \quad \alpha+\beta = 1. \tag{5}$$

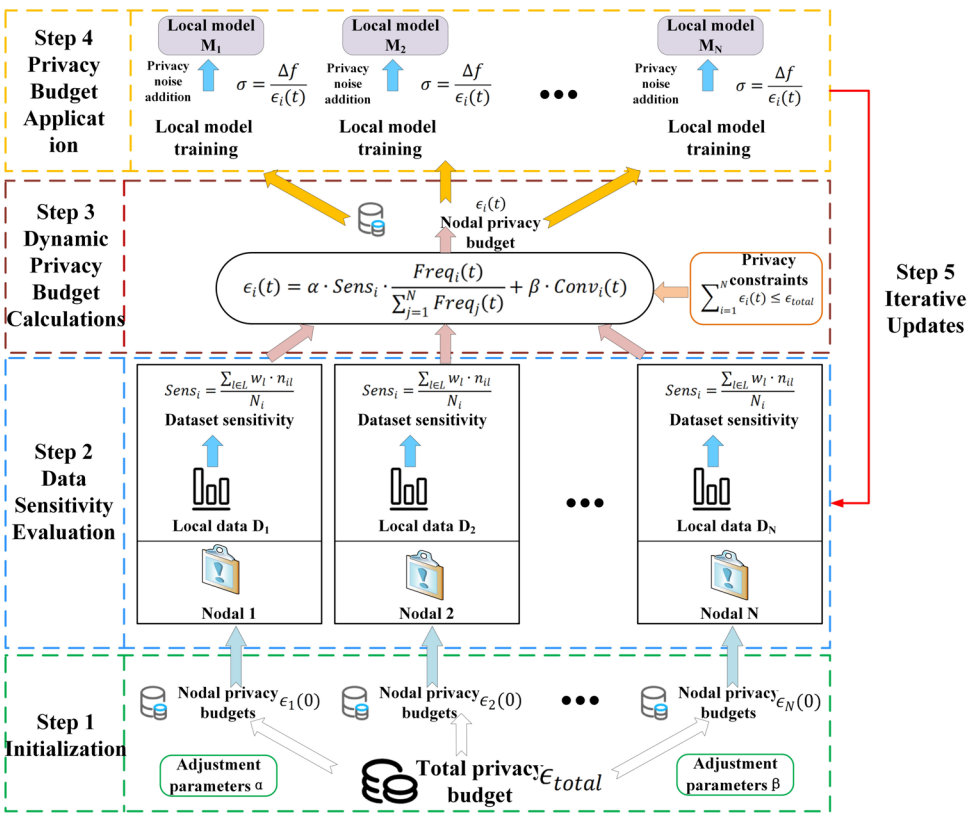


Fig. 1 Dynamic Privacy Budget Allocation Algorithm Framework Diagram

Here, Sens_i quantifies the inherent privacy sensitivity of D_i, $\mathrm{Freq}_i(t)$ is the relative number of times U_i participated up to round t, and $\mathrm{Conv}_i(t)$ measures the past impact of the node on the reduction of global loss. By combining these factors, the allocation of the budget is dynamically adapted to the characteristics of the data and historical contributions of each node, while the global constraint $\sum_i \epsilon_i(t+1) \leq \epsilon_{\text{total}}$ is enforced by proportionally scaling down any overbudget assignments. This interaction model thus guarantees rigorous differential privacy for all participants and optimizes the trade-off between privacy protection and learning performance.

Our system model for DPBA is organized into five main steps: (1) the server initializes and allocates the global privacy budget ϵ_{total} among all N nodes; (2) each node U_i receives the global model $W^{(t)}$ and its assigned budget $\epsilon_i(t)$; (3) U_i performs a differentially-private local update by injecting Laplace noise proportional to $\Delta f/\epsilon_i(t)$; (4) the server collects and securely aggregates all noisy updates via weighted averaging to form $W^{(t+1)}$; and (5) the server dynamically recalculates each node's next-round budget $\epsilon_i(t+1)$ based on data sensitivity Sens_i, participation frequency $\mathrm{Freq}_i(t)$, and convergence contribution $\mathrm{Conv}_i(t)$, enforcing $\sum_i \epsilon_i(t+1) \leq \epsilon_{\text{total}}$. This five-step workflow ensures adaptive privacy protection aligned with node characteristics while maintaining rigorous differential privacy. Next, we will follow these 5 steps to describe the system model.

3.2.2 Initialization

In this stage, the total privacy budget ϵ_{total} available for the entire FL process is set as the upper limit to protect privacy within the system. Subsequently, each node i is initialized with its initial privacy budget $\epsilon_i(0)$, ensuring that every node possesses a basic level of privacy protection at the outset of FL. In addition, the adjustment parameters α and β are defined to flexibly control the allocation of privacy budgets. These parameters adjust the weighting of the sensitivity scores relative to other critical factors (such as the frequency of node participation and model convergence) in the privacy budget distribution. By setting these initial parameters appropriately, the system can effectively manage privacy budgets throughout the learning process, achieving a balance between privacy protection and model performance.

3.2.3 Data sensitivity assessment

For each node i, the sensitivity Sens_i of its data set D_i is assessed to determine the privacy protection requirements for the data from that node. The sensitivity assessment can be based on factors such as the type of data and the level of sensitivity of the content. Assume that there are three levels of sensitivity l_H, l_M, and l_L, representing high, medium, and low sensitivity, with the corresponding weights w_H, w_M, and w_L, respectively. The sensitivity score Sens_i of the data set of node i D_i can be calculated using the following formula:

$$\mathrm{Sens}_i = \frac{\sum_{l \in L} w_l \cdot n_{il}}{N_i}, \tag{6}$$

where n_{il} is the number of data elements in the dataset D_i with sensitivity level l, N_i is the total number of data elements in D_i, and w_l is the weight corresponding to sensitivity level l, with $w_l \in \{w_H, w_M, w_L\}$.

This formula provides a weighted average that considers different levels of sensitivity within the data set to calculate the overall sensitivity score for node i. This score reflects the overall sensitivity level of the node's data and serves as a basis for subsequent privacy budget allocation, ensuring that more sensitive data receive stronger privacy protection.

3.2.4 Dynamic privacy budget calculation

At each time step t, the dynamic privacy budget $\epsilon_i(t)$ for each node i is calculated as shown in Eq. (5). The parameter α controls the weight of the data sensitivity and participation frequency, while β determines the impact of the node's contribution to model convergence. These parameters ensure flexibility in the allocation of privacy budgets, allowing the system to dynamically adapt to the varying characteristics of each node.

The formula incorporates the participation frequency $\mathrm{Freq}_i(t)$, which reflects the relative activity of each node, ensuring that the nodes that participate more frequently receive proportionally higher privacy budgets. Similarly, the contribution to the convergence of the model $\mathrm{Conv}_i(t)$ captures the influence of updates from a node on improving the performance of the global model. This dual consideration allows the system to prioritize nodes that are both active and impactful, achieving a balance between privacy preservation and overall model efficiency.

The privacy budget constraint ensures that the sum of privacy budgets for all nodes does not exceed the total privacy budget, as shown in Eq. (1).

This constraint is crucial to maintaining the privacy guarantees of the overall system. If the total sum of the allocated privacy budgets exceeds ϵ_{total}, the system applies a proportional reduction to the $\epsilon_i(t)$ of each node. Specifically, each node's budget is scaled down uniformly to ensure that the total allocation remains within the defined limit. This proportional adjustment maintains the relative differences in privacy budgets between nodes, preserving the prioritization established by the original calculation.

The dynamic adjustment of $\epsilon_i(t)$ through this approach ensures that nodes with higher sensitivity, higher participation frequency, or higher contributions to convergence are allocated

higher privacy budgets. Meanwhile, the proportional reduction mechanism guarantees compliance with the total privacy budget constraint, allowing the system to maintain robust DP guarantees while optimizing resource utilization across the FL network.

3.2.5 Privacy budget application

The calculated privacy budget $\epsilon_i(t)$ is applied during the local model training and update processes for node i to protect its data privacy. DPBA applies the privacy budget $\epsilon_i(t)$ to each node i's local convolutional neural network (CNN) model training and update process to ensure data privacy protection. For each node i, the DP noise is calculated in each update of the model. The noise magnitude σ can be calculated based on the privacy budget $\epsilon_i(t)$:

$$\sigma = \frac{\Delta f}{\epsilon_i(t)}, \tag{7}$$

where Δf is the sensitivity of the query function. After each local model training, noise is applied to the model's gradients or weight updates. Let g represent the gradient without noise; the gradient with noise is $g + \text{Laplace}(0, \lambda)$.

The local model employs a convolutional neural network (CNN) consisting of multiple convolutional layers and fully connected layers. The basic structure is Convolution $\rightarrow$ ReLU $\rightarrow$ Pooling $\rightarrow$ Fully Connected. The formula for the convolutional layer is the following:

$$f(x) = W * x + b, \tag{8}$$

where W is the convolution kernel, $*$ denotes the convolution operation, x is the input and b is the bias term. During training of the local CNN model, parameters such as weights W and bias b are updated as follows:

$$W = W - \eta(\nabla W + \text{noise}), \tag{9}$$

$$b = b - \eta(\nabla b + \text{noise}), \tag{10}$$

where η is the learning rate and ∇W, ∇b are the corresponding gradients.

3.2.6 Iteration and update

In global model aggregation, each participating node trains its local model on its own dataset, calculating updates for model parameters (such as weights and biases). At the end of each iteration, the FL system aggregates the local model updates from each node and uses the federated averaging method to update the global model. The formula for the global model update is as shown in Eq. (4).

After each iteration, the system assesses the convergence status of the global model. The validation set can be used to calculate the change in loss or precision of the global model. A significant reduction in loss or an improvement in accuracy indicates that the model is converging; if the model performance changes minimally, it suggests that the model is nearing convergence. Based on the current state of convergence of the global model and the participation frequency of the nodes, the privacy budget of each node $\epsilon_i(t+1)$ is recalculated.

If the total exceeds ϵ_{total}, each node's privacy budget is proportionally reduced to ensure the sum remains within the defined range. Each node then retrains its local model based on the allocated privacy budget $\epsilon_i(t+1)$. During local training, the noise level is adjusted according to the new privacy budget to ensure DP protection. After local model training, each node calculates its model update and uploads it again to the central server for global model updates.

To address the problem that the effectiveness of the DPBA algorithm is highly dependent on an accurate sensitivity assessment of the data at each node,, we introduce a hybrid quantitative sensitivity scoring mechanism that combines statistical characteristics -such as the data distribution deviation δ_i and the privacy risk indicator ρ_i - with annotations from experts in the domain ℓ_i. The sensitivity score for node *i* is computed as

$$S_i = h(\delta_i, \rho_i, \ell_i; \theta). \tag{11}$$

The parameters θ of the sensitivity assessment function $h(\cdot)$ are then automatically optimized through cross-validation and calibration *K* times in a small set of labeled samples $\mathcal{L}$.

On the other hand, to prevent allocation bias caused by assessment errors, an adaptive correction module is integrated into the algorithm. Denote the historical sensitivity score of node *i* by $\bar{S}_i$ and the actual measured risk of privacy leakage by R_i. Whenever their absolute deviation exceeds a threshold τ,

$$\left|\bar{S}_i - R_i\right| > \tau. \tag{12}$$

The budget allocation weight $\omega_i(t)$ is dynamically corrected as

$$\omega_i'(t) = \omega_i(t)\frac{R_i}{\bar{S}_i}, \tag{13}$$

thereby ensuring that low-sensitivity data are not overprotected and high-sensitivity data are not underprotected.

The steps from Step 3 to Step 5 are repeated continuously until the preset number of iterations is reached or until the global model converges. In each iteration, the allocation of the privacy budget is dynamically adjusted to meet the privacy needs of each node while maximizing the efficiency of training and the performance of the FL system model.

3.2.7 DPBA algorithm and pseudo-code

The pseudocode of the DPBA Algorithm is shown in Algorithm 1. Specifically, the DPBA algorithm operates through five main steps to effectively manage privacy in a FL system. **First, Initialization** sets the stage by defining the total privacy budget, ϵ_{total}, and assigning the initial privacy budgets, $\epsilon_i(0)$, to each of the N nodes, ensuring that the sum of individual budgets does not exceed ϵ_{total}. **Second, Data Sensitivity Assessment** evaluates the sensitivity of each node's dataset, D_i, assigning a sensitivity score, Sens_i, based on predefined sensitivity levels. **Third, Dynamic Privacy Budget Calculation** dynamically adjusts the privacy budget $\epsilon_i(t)$ for each node at each iteration t considering factors such as data sensitivity, participation frequency $\text{Freq}_i(t)$, and the node's contribution to model convergence $\text{Conv}_i(t)$. If the total budget allocated exceeds ϵ_{total}, the algorithm proportionally reduces each $\epsilon_i(t)$ to maintain the budget constraint. **Fourth, Privacy Budget Application** involves applying the calculated privacy budget $\epsilon_i(t)$ to each node's local model training process by injecting noise into the model updates, ensuring DP. **Finally, Iteration and Update** aggregates the locally trained models into a global model using methods such as weighted averaging, assesses the convergence of the global model and repeats the process until convergence is achieved or the maximum number of iterations T is reached. Throughout these steps, the DPBA algorithm dynamically balances privacy protection with model performance by intelligently allocating the privacy budget based on each node's characteristics and contributions.

Algorithm 1 DPBA Algorithm

Require: • Number of nodes: N
- Datasets: $\{D_i\}_{i=1}^{N}$
- Total privacy budget: ϵ_{total}
- Initial budgets: $\{\epsilon_i(0)\}_{i=1}^{N}$
- Tuning parameters: α and β (with $\alpha + \beta = 1$)
- Maximum iterations: T

Ensure: • A converged global model $W^{(t)}$ with effective privacy guarantees

Step 1: Initialization
Set the total privacy budget ϵ_{total}.
Initialize each node i with $\epsilon_i(0)$, ensuring $\sum_{i=1}^{N} \epsilon_i(0) \leq \epsilon_{\text{total}}$.
Set or tune α and β.
Step 2: Data Sensitivity Assessment
for $i \leftarrow 1$ to N **do**
    Evaluate data sensitivity of node i (e.g., via sensitivity level counts).
    Store as Sens_i.
end for
for $t \leftarrow 0$ to $T-1$ **do**
    Step 3: Dynamic Privacy Budget Calculation
    for $i \leftarrow 1$ to N **do**
        Obtain node i's participation frequency $\text{Freq}_i(t)$.
        Obtain node i's convergence contribution $\text{Conv}_i(t)$.
        Update $\epsilon_i(t)$ using:
    $\epsilon_i(t) \leftarrow \alpha \times \text{Sens}_i + \beta \times \left[\text{Freq}_i(t) \text{ and } \text{Conv}_i(t)\right]$.
    end for
    If $\sum_{i=1}^{N} \epsilon_i(t) > \epsilon_{\text{total}}$, reduce each $\epsilon_i(t)$ proportionally.
    Step 4: Privacy Budget Application
    for $i \leftarrow 1$ to N **(in parallel) do**
        Inject differentially private noise into local updates based on $\epsilon_i(t)$.
        Train local model using noisy gradients.
    end for
    Step 5: Iteration and Update
    Aggregate local models to form global model $W^{(t+1)}$.
    Check convergence criteria (loss/accuracy). If converged, **break**.
end for
Output: Final global model $W^{(t)}$ (after convergence or T iterations).

4 Experimental and analysis

4.1 Datasets

To validate the effectiveness of the DPBA algorithm, we use two publicly available benchmark datasets. MNIST and CIFAR-10, which are widely used in machine learning and computer vision tasks.

(1) The MNIST [30] (Modified National Institute of Standards and Technology) dataset is one of the most well known datasets in the field of machine learning, commonly used to evaluate algorithms in the context of image classification. It consists of grayscale images of handwritten digits, ranging from 0 to 9, totaling 10 distinct categories. Each image in the dataset has a fixed size of 28×28 pixels, and it contains 60,000 training samples and 10,000 test samples. The dataset has been widely studied and serves as a standard benchmark for evaluating the performance of various machine learning models, particularly in tasks such as digit recognition and general image classification.

(2) The CIFAR-10 dataset [31] (Canadian Institute of Advanced Research) is another well-established benchmark in the field of machine learning. It contains 60,000 32×32 color images, which are categorized into 10 different classes. Each class has 6,000 images, resulting in a total of 60,000 images. The classes in this dataset include a wide range of everyday objects, such as airplanes, cars, birds, cats, dogs, frogs, horses, ships, trucks, and more. The CIFAR-10 dataset is divided into 50,000 training images and 10,000 test images and is often used to test the performance of image classification models, especially those involving convolutional neural networks (CNN). The relatively small size of the images and the diversity of object categories make CIFAR-10 a challenging dataset for model evaluation.

Both datasets are chosen because they represent different levels of difficulty and data characteristics. Although MNIST is relatively simple due to the small image size and clear category distinctions, CIFAR-10 is more complex, with larger, color images, and more varied object categories. These data sets provide a comprehensive testbed for validating the DPBA algorithm, as they allow the evaluation of model performance across both simple and more complex image classification tasks.

4.2 Experimental setup

The experimental environment consists of a CPU with an Intel Core i7-9700K 3.60 GHz processor, an NVIDIA GeForce RTX 3080 with 12 GB of VRAM, and 16 GB of RAM. The experiments are conducted using the PyTorch 1.6 deep learning framework and Python 3.7. The MNIST and CIFAR-10 datasets are evenly distributed in 100 simulated nodes, creating an FL environment. Each node represents an independent training participant, and each node trains a shared model using its local data, periodically uploading model updates. The initial privacy budget, ϵ_{total}, is set at a fixed value of 2.0. The parameters of the DPBA algorithm are set as $\alpha = 0.6$ and $\beta = 0.4$, which are adjusted to balance the expected privacy protection and the performance of the model. The sensitivity score (Sens_i) is set to 1.0. A convolutional neural network (CNN) model is used for both the MNIST and CIFAR-10 datasets, with training parameters including a learning rate of 0.0001 and a batch size of 128. The model is trained for 50 epochs, which is sufficient to observe the learning effect.

4.3 Comparison methods

In order to verify the validity of the method proposed in this paper, the following comparison methods are designed.

(1) **DPSaab**. A DP subspace approximation algorithm with adjusted bias (DPSaab) was proposed by Wang et al. [28]. to protect training data in FF-CNN. The allocation of privacy budgets is designed according to the contribution of model filters to the output response, allowing the allocation of larger privacy budgets to filters with greater contributions, based on the ratio of feature values.

(2) **A-DPFL**. Zhao et al.[29] proposed a novel federated privacy-based learning framework. They designed an adaptive differentiated privacy FL algorithm (A-DPFL) and a multilevel multiparticipant dynamic privacy budget allocation method to reduce injected noise, effectively improving the utility.

(3) **BRPPNet**. Lin et al. [32] proposed a lightweight but effective protection network, Balanced Referring Personal PrivacyNet (BRPPNet), which introduced a multi-scale feature fusion module and appropriate balanced-BCE loss.

(4) **EtC**. Nakachi et al. [33] proposed lightweight scrambled JPEG XS encoding method to preserve privacy. The proposed method is an encryption followed by compression (EtC) system. It enables the use of EtC schemes while maintaining compatibility with JPEG XS.

(5) **DPBA-statistic**. This method follows the framework architecture proposed in this paper, but uses a fixed privacy budget allocation, based on the experimental setup in Sect. 4.2.

(6) **DPBA**. The DPBA proposed in this paper aims to optimize the application of DP in FL environments, following the experimental setup in Sect. 4.2.

4.4 Evaluation criteria

This paper uses four key metrics as evaluation criteria to comprehensively assess the effectiveness of DP mechanisms in FL, particularly the DPBA algorithm. We assume that the global model is f_θ, the training set is $\mathcal{D}_{\text{train}}$, and the test set is $\mathcal{D}_{\text{test}}$, where each sample is (x_i, y_i).

(1) **Accuracy** [34]: This metric is used to measure the classification performance of the FL model in the test dataset. It reflects the prediction accuracy of the model for a given task and represents the overall performance of the model.

$$\text{Accuracy} = \frac{\sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = y_i)}{N}, \tag{14}$$

where N is the total number of samples, $\hat{y}_i$ and y_i are the predicted and true labels of the i-th sample, respectively, and $\mathbb{I}(\cdot)$ is the indicator function, which equals 1 if the inside condition is true and 0 otherwise.

(2) **Privacy Leakage Rate (PLR)** [35]: This metric is used to assess the likelihood of personal data leak during the FL process. The lower the privacy leakage rate, the stronger the privacy protection. In this paper, we calculate the privacy leakage risk of each method on different datasets through simulated attacks (e.g., reconstruction attacks).

A reconstruction attack algorithm $\mathcal{A}_{\text{rec}}$ is constructed to attempt to reconstruct each of the N_{test} samples in $\mathcal{D}_{\text{test}}$. The number of samples successfully reconstructed is denoted by N_{rec}. The privacy leakage rate is then calculated as

$$\text{PLR} = \frac{N_{\text{rec}}}{N_{\text{test}}}. \tag{15}$$

(3) **Model Inference Attack (MIA) Resistance** [36]: MIA resistance is used to measure the model's ability to withstand inference attacks. Inference attacks typically attempt to infer private information from the model's output, and the higher the MIA resistance percentage, the stronger the model's privacy protection ability.

An inference attack model $\mathcal{A}_{\text{mia}}$ is used. From the member set $\mathcal{D}_{\text{train}}$, N_{mem} examples are sampled, and from the non–member set $\mathcal{D}_{\text{test}}$, N_{non} examples are sampled. For each example, the attack model produces an output $\hat{m}_i \in \{0, 1\}$, where 1 indicates prediction as a member. The inference accuracy is then computed as

$$\text{MIA} = \frac{1}{N_{\text{mem}} + N_{\text{non}}} \sum_{i=1}^{N_{\text{mem}}+N_{\text{non}}} \mathbb{I}(\hat{m}_i = m_i), \tag{16}$$

where m_i denotes the true membership label.

(4) **Adversarial Attack Success (AAS) Rate** [37]: This metric is used to assess the model's vulnerability to adversarial inputs. The lower the adversarial attack success rate, the stronger the model's resistance to adversarial attacks and the higher its security.

For each test sample x_i, an adversarial example $x_i' = x_i + \delta$ is generated with $\|\delta\| \leq \epsilon$. The model prediction for the adversarial example is denoted by $\hat{y}_i = f_\theta(x_i')$. The number of adversarial examples that are misclassified is indicated by N_{adv}. The adversarial attack success rate is then calculated as

$$\text{AAS} = \frac{N_{\text{adv}}}{N_{\text{test}}}. \tag{17}$$

4.5 Experimental results and analysis

The experimental results of different comparison methods on the MNIST and CIFAR-10 datasets are shown in Table 1. From Table 1, the following conclusions can be drawn:

The experimental results in Table 1 clearly highlight the effectiveness of the DPBA method in both the MNIST and CIFAR-10 datasets. In the MNIST and CIFAR-10 datasets, DPBA achieved the highest accuracy of 96.6% and 90.2%, significantly exceeding all other methods. The strong performance of DPBA suggests that it strikes a superior balance between model accuracy and privacy protection compared to the other approaches tested. Its ability to deliver the highest accuracy while maintaining strong privacy guarantees sets it apart from the competition. DPBA also demonstrated the lowest PLR (1.4, 3.1) and AAS (19.7, 15.1), further highlighting its ability to strike an optimal balance between privacy protection and accuracy. These results across both datasets consistently show that DPBA outperforms existing methods in both model accuracy and privacy-preserving capabilities.

Table 1 Experimental results of different comparison methods on MNIST dataset and CIFAR-10 dataset

Dataset	Method	Accuracy	PLR	MIA	AAS
MNIST	DPSaab	93.8	2.3	84.2	55.1
	A-DPFL	94.2	2.1	85.2	23.9
	BRPPNet	92.7	3.5	87.0	27.8
	EtC	94.1	2.0	88.4	21.9
	DPBA-statistic	93.1	3.0	88.3	26.4
	DPBA	**96.6**	**1.4**	**89.6**	**19.7**
CIFAR-10	DPSaab	86.5	5.7	82.2	60.7
	A-DPFL	84.1	7.9	79.3	36.9
	BRPPNet	83.5	6.8	73.1	33.7
	EtC	86.2	5.0	75.9	24.4
	DPBA-statistic	84.4	6.2	70.5	14.8
	DPBA	**90.2**	**3.1**	**85.8**	**15.1**

The bolded numbers represent the optimal values under different evaluation criteria

When assessing MIA results, DPBA demonstrated its robustness against attacks by achieving the highest MIA score of 89.6 in MNIST, 85.8 in CIFAR-10. This means that DPBA is highly resistant to membership inference attacks that attempt to determine whether a specific data point was included in the training data. Other methods, such as DPSaab (84.2, 82.2) and BRPPNet (87.0, 73.1), had higher MIA values, suggesting that they are more vulnerable to such attacks and therefore less secure in terms of privacy preservation. The superior MIA performance of DPBA underscores its effectiveness in protecting sensitive data while maintaining model accuracy.

In terms of PLR, DPBA once again excelled, with the lowest value of 1.4 in MNIST, 3.1 in CIFAR-10. This indicates that DPBA effectively minimizes privacy leakage, a critical requirement in privacy-preserving machine learning systems. In comparison, DPSaab (2.3, 5.7) and EtC (2.0, 5.0) showed relatively higher PLR values, signaling that these methods may suffer more loss of privacy. Both A-DFPL and BRPPNet recorded higher PLR values, further suggesting that these approaches trade off some level of privacy in favor of model performance.

In terms of AAS, which evaluates the trade-off between privacy protection and model accuracy, DPBA again achieved the best result with the lowest AAS of 19.7 in MNIST and 15.1 in CIFAR-10. This reflects its ability to provide the best balance between maintaining high accuracy and strong privacy protection.

The DPBA method outperforms the DPBA-statistic method in both the MNIST and CIFAR-10 datasets in terms of accuracy, privacy protection, and defense against attacks. In MNIST, DPBA achieved a higher accuracy of 96.6%, compared to DPBA-statistic's 93.1%, showing that DPBA provides better model performance. Similarly, DPBA showed lower scores for PLR (1.4) and MIA (19.7), indicating superior preservation of privacy. In contrast, the DPBA statistic had higher PLR (3.0) and MIA (26.4) values, suggesting that it is more susceptible to privacy leakage and membership inference attacks. Overall, DPBA shows clear advantages over the DPBA-statistic in terms of both model performance and privacy protection, making it the most effective method for FL applications.

We quantify the additional overhead of the dynamic tuning process through theoretical analysis and experimental measurements and compare it with traditional fixed-budget methods. The results show that in a typical scenario with 100 nodes and 50 rounds of iterations, the total time overhead of DPBA increases only by about 8%, while the model accuracy improves by more than 3.5% and 5.8%. To address this situation, we propose two optimization strategies: one is to asynchronously parallelize parameter updates with global aggregation to reduce computational delays, and the other is to employ a sparse trigger mechanism to recompute the budget only when the node sensitivities or convergence contributions are significantly changed, thus keeping the additional overhead at a minimum level. In the discussion, we explicitly point out that the complexity can be further reduced in the future by lightweight parameter estimation and hierarchical budget adjustment to realize the feasibility of large-scale deployment.

In a typical experiment with 100 nodes and 50 rounds of iterations, the computational overhead of DPBA increases only by approximately 8%, and the communication overhead increases by approximately 5%. Meanwhile, we propose and evaluate three optimization strategies: first, asynchronous parallel processing is used between budget computation and global aggregation to hide the delay; second, a sparse trigger mechanism is introduced to recalculate the budget only when the node sensitivities or convergence contributions change significantly; and third, local approximation and hierarchical budget computation are used to significantly reduce the real-time adjustment frequency while guaranteeing the accuracy. Experimental results show that after the above optimizations, the system can still maintain good performance in large-scale deployment scenarios, and supports further combining edge computing and communication compression techniques to enhance scalability.

4.6 Comparative analysis of experimental parameters

4.6.1 Impact of tuning parameter α

The effect of varying the tuning parameter α on model accuracy across the MNIST and CIFAR-10 datasets is shown in Fig. 2. The experimental results indicate that α significantly influences the performance of the models in both data sets, although the trends differ between them. For the MNIST dataset, the accuracy initially increases as α increases, reaching a peak at $\alpha = 0.6$ with an accuracy of 96.6%. After this point, further increases in α lead to a decrease in performance. This trend suggests that an optimal value α is crucial to balance sensitivity and convergence contributions, thus maximizing accuracy for simpler datasets such as MNIST. A smaller α underutilizes the sensitivity information, while a larger α may overuse it, leading to suboptimal results. This observation underscores the importance of carefully tuning α to achieve the best performance of the model.

For the CIFAR-10 dataset, the accuracy shows a relatively steady upward trend as α increases, with the highest accuracy values of 90.4% and 88.6% achieved at $\alpha = 0.7$ and $\alpha = 0.8$, respectively. However, at $\alpha = 0.9$, a slight drop in accuracy is observed, which highlights the sensitivity of the model to too high values α. Unlike MNIST, where

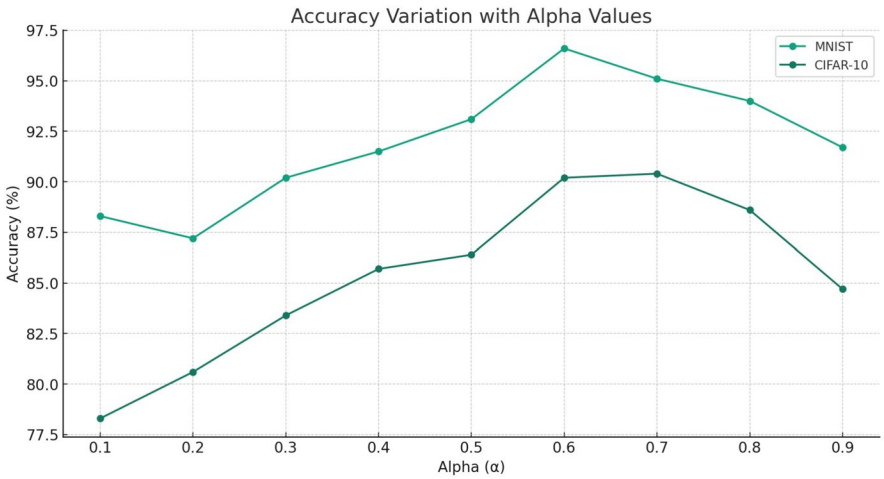


Fig. 2 Changes in accuracy of MNIST and CIFAR-10 datasets with different weights $\alpha = 0.1 - 0.9$

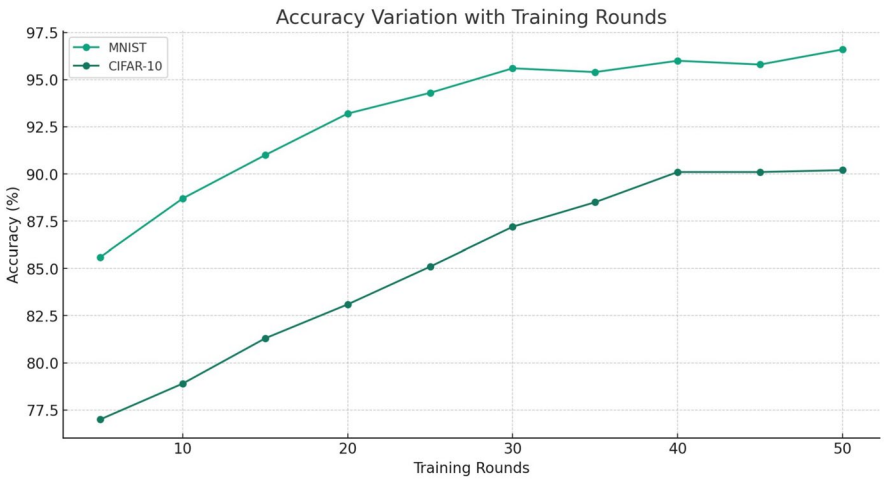


Fig. 3 Plot of change in accuracy for MNIST and CIFAR-10 datasets with different number of training rounds

the accuracy starts to decline at much lower α, the CIFAR-10 dataset benefits from higher α, possibly due to its greater complexity and the greater impact of sensitivity-aware contributions. These findings emphasize that the choice of α is dependent on the data set and critical to balancing privacy protection and model accuracy. A well-chosen α ensures sufficient privacy preservation without compromising learning efficiency, which is particularly important for more complex datasets such as CIFAR-10.

4.6.2 Comparison of results at different numbers of training rounds

The variation in model accuracy under different training rounds is illustrated in Fig. 3. For the MNIST dataset, the accuracy shows a clear upward trend as the number of training rounds increases. During the early stages of training (5-20 rounds), the accuracy increases dramatically, indicating that the model quickly adapts to the relatively simple patterns in the dataset. This rapid improvement suggests that initial rounds are crucial for capturing the fundamental features of MNIST, enabling the model to achieve substantial gains in accuracy within a short timeframe. After 20 training rounds, the rate of improvement begins to slow as the model refines its learning. By the time the training reaches 40 rounds, the accuracy stabilizes at a high value, suggesting that the model has effectively converged for this dataset. The results imply that, for datasets with relatively low complexity, such as MNIST, fewer training rounds may be sufficient to achieve optimal model performance.

In contrast, the accuracy of the CIFAR-10 data set improves more gradually with additional training rounds, reflecting its greater complexity and the model's need for more iterations to capture intricate features. Although precision increases steadily throughout the training process, early stage improvements (5–20 rounds) are less pronounced compared to MNIST, as the model requires additional rounds to adapt to the diverse and challenging characteristics of CIFAR-10. Between 35 and 50 training rounds, the accuracy reaches a plateau, indicating that the model has nearly converged and further improvements are minimal. These findings underscore the importance of tailoring the number of training rounds to the complexity of the dataset. While simpler datasets such as MNIST require

fewer iterations for convergence, complex datasets such as CIFAR-10 demand more training to achieve similar levels of accuracy. Thus, selecting an appropriate number of training rounds is essential to ensure computational efficiency while maximizing model performance in FL systems.

4.7 Discussions

The proposed DPBA algorithm presents a novel approach to managing privacy in FL systems. By dynamically adjusting the privacy budget allocated to each node based on key factors such as data sensitivity, participation frequency, and node's contribution to model convergence, DPBA addresses the inherent limitations of traditional methods that use a fixed privacy budget. The dynamic allocation mechanism allows the system to prioritize nodes with more sensitive data and those that participate more frequently, ensuring that these nodes receive adequate privacy protection. At the same time, it allows active nodes with significant contributions to the convergence of the model to continue participating in the learning process without excessive loss of privacy. This balance between privacy protection and model efficiency represents a major improvement over static privacy budget allocation methods, which are often unable to adapt to the diverse characteristics of different nodes in FL environments.

Despite its advantages, there are some challenges and limitations that the DPBA algorithm faces. One key issue is the reliance on a global privacy budget constraint, which ensures that the sum of all allocated privacy budgets does not exceed the total available privacy budget. Although this approach guarantees overall system-wide privacy protection, it can limit the ability to allocate sufficient privacy budgets to nodes that are highly sensitive or particularly active in the learning process. In practice, some nodes may require significantly more privacy protection than others due to differences in data sensitivity or participation frequency. In such cases, the global constraint could lead to an inequitable distribution of privacy budgets, potentially leaving highly sensitive nodes with inadequate privacy protection. A more flexible and decentralized approach to privacy budget allocation could address this limitation by considering the varying needs of individual nodes and dynamically adjusting the total budget at the node level, thus enabling more equitable and adaptive privacy protection across the system.

Another potential limitation of the DPBA algorithm is its reliance on the accuracy of the sensitivity assessment and the model convergence metrics. The effectiveness of the algorithm depends on how well these metrics are calculated. If the sensitivity of data from a node is incorrectly assessed or if the convergence of the global model is not properly measured, the algorithm could allocate privacy budgets inefficiently. This could result in some nodes receiving too little privacy protection or, conversely, overallocating privacy budgets to nodes that do not require as much. To improve the effectiveness of the DPBA algorithm, more research is needed to enhance the precision and robustness of sensitivity evaluation methods and convergence detection techniques. Additionally, as the FL system scales up and the number of participating nodes increases, the computational complexity of dynamically adjusting privacy budgets becomes a significant consideration. The algorithm's efficiency in real-time adjustments must be optimized to handle large-scale dynamic datasets without introducing delays or excessive computational overhead.

In this paper, the main assumptions are that the data distribution of each node is relatively stable and the sensitivity can be quantified, the measurement method to simplify the contribution to the convergence of the global model, and the settings of the fixed hyperparameters. Additionally, the influence of these assumptions on the algorithm performance under non-IID data, different noise distributions, and parameter variations is evaluated through additional ablation experiments. The experimental results show that although simplification improves implementability, DPBA can still maintain a good balance of privacy and utility in more complex scenarios. We also clarified the limitations of these assumptions in the conclusion and proposed the future working direction: in a more lenient non-IID environment, dynamically optimize the hyperparameters and flexibly model the convergence contribution to further improve the universality and robustness of DPBA.

We also recognize the potential risks that the participation of high-frequency malicious nodes can pose to system security and privacy protection. On the one hand, such nodes can utilize their frequent model update requests to repeatedly detect the differences between the noise and the true gradient in the global model and thereby infer sensitive information. However, by accumulating small but targeted disturbances, they have the potential to gradually undermine model performance or bypass privacy budget constraints. Therefore, future research will focus on exploring frequency anomaly detection and hierarchical aggregation strategies based on node behavior patterns. By dynamically monitoring the degree of deviation between local updates and global trends, suspicious nodes can be marked and restricted in a timely manner. Meanwhile, dynamic reputation scoring and multiparty security verification mechanisms are introduced to balance efficiency and credibility; and these strategies will be considered to be combined with homomorphic encryption or secure multi-party computation to further enhance the robustness and privacy protection capabilities of the DPBA framework in large-scale heterogeneous environments.

Future research could expand the application of the DPBA algorithm to more complex domains, such as healthcare and finance, and optimize its adaptability and computational efficiency, particularly in large-scale, distributed environments. Investigating the combination of DPBA with other privacy-preserving techniques, such as secure multiparty computing (SMPC) and homomorphic encryption, could enhance its effectiveness overall. Analyzing the impact of DPBA on communication costs and computational resource consumption in real-world applications is crucial for the implementation of the algorithm in industrial settings.

5 Conclusion and future work

In this paper, we propose a DPBA algorithm to overcome the limitations of traditional fixed privacy budget allocation methods in FL. DPBA dynamically adjusts the privacy budget based on factors such as node data sensitivity, participation frequency, and model convergence state. This approach achieves an optimal balance between privacy protection and model performance. Experimental results show that DPBA significantly improves model accuracy and reduces privacy leakage rate. DPBA also improves the resistance to inference and adversarial attacks in various datasets, including MNIST and CIFAR-10. This research introduces novel approaches to the application of DP in FL and highlights the potential of DPBA to enhance the security and efficiency of FL systems.

Future research could expand the application of the DPBA algorithm to more complex domains, such as healthcare and finance, and optimize its adaptability and computational efficiency, particularly in large-scale, distributed environments. Investigating the combination of DPBA with other privacy-preserving techniques, such as SMPC and homomorphic encryption, could enhance its effectiveness overall. Analyzing the impact of DPBA on communication costs and computational resource consumption in real-world applications is crucial for deploying the algorithm in industrial-scale settings.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest We declare that the authors have no competing interests as defined by Springer, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

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